

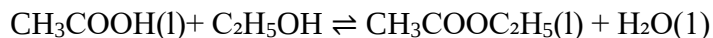
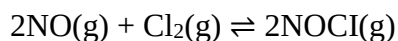
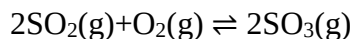
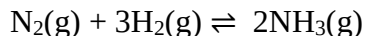
CHEMICAL EQUILIBRIUM

TYPES OF EQUILIBRIUM

1. Homogeneous Equilibria

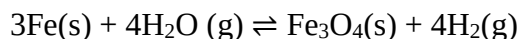
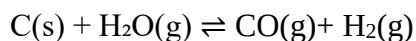
An equilibrium system in which all of the reactants and products are in the same phase.

For example



2. Heterogeneous Equilibria

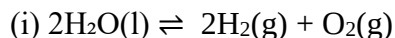
Equilibria which involve more than one phases are called Heterogeneous equilibria. For example



For pure substance

If pure solids or pure liquids are involved in an equilibrium system, their concentrations are not included in the equilibrium constant expression.

This is because the change in concentration of any pure solid or liquid has no effect on the equilibrium system.



$$K_c = [\text{H}]^2 [\text{O}_2]$$

and

$$K_p = P_{\text{H}_2} P_{\text{O}_2}$$

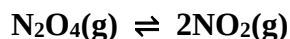
Way to Recognize Equilibrium

a) Physical Method (spectrometric method)

This method is based on **measuring the physical properties** of the reaction mixture.

This method can be applied when the reactant or product absorbs ultraviolet, visible or infrared radiation.

The concentration can be determined by measuring the amount of radiation absorbed.

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(Colourless) (Reddish brown)

N_2O , is a colourless gas whereas NO_2 is reddish brown gas. The progress of the reaction can be studied by **measuring the absorbance** at regular Intervals. **Absorbance $\propto$ NO_2 .**

At equilibrium spectrometer will show constant value of absorbance.

b) Chemical Method

- ✓ In this method, the amount of a reactant or product is determined by a suitable chemical reaction.
- ✓ Consider the reaction between acetic acid and ethanol to form ethyl acetate and water.

$$\text{CH}_3\text{COOH}(\text{l}) + \text{C}_2\text{H}_5\text{OH}(\text{l}) \rightleftharpoons \text{CH}_3\text{COOC}_2\text{H}_5(\text{l}) + \text{H}_2\text{O}(\text{l})$$
- ✓ The progress of the reaction can be studied by determining the concentration of acetic acid after regular intervals.
- ✓ For this purpose, a small portion of mixture is withdrawn. The concentration of acetic acid is determined by titrating it against a standard solution of NaOH using **phenolphthalein** as an indicator.
- ✓ The concentration of acetic acid will decrease until equilibrium is attained. At equilibrium the concentration of acetic acid will become constant, indicating that equilibrium has been reached.

THE LE-CHATelier'S PRINCIPLE

It can be stated as

"If a change is imposed on a system at equilibrium, the position of the equilibrium will shift in a direction which tends to reduce that change".

Or

"If you impose a change on a system, the system will act so as to nullify the change"

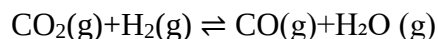
Or

"If a stress is applied on a system in equilibrium, the position of the equilibrium shift in the direction which reduces the stress."

FACTORS AFFECTING EQUILIBRIUM**1. Effect of change in concentration:**

When the concentration of one or more of the reactants or products present in the equilibrium mixture is disturbed, the system will not remain in the equilibrium state. According to Le Chatelier's principle, the equilibrium shifts to accommodate the substance added or removed and restore equilibrium again.

Consider the following gas phase equilibrium:

**Increasing CO_2 concentration**

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When CO₂ is added to this equilibrium system, it is no longer in equilibrium.

Higher concentration of CO₂ increases the rate of forward reaction relative to the reverse reaction. Thus more CO₂ and H₂ combine and more CO and H₂O are formed.

Decreasing CO₂ and H₂ concentration

As time passes the concentrations of CO₂ and H₂ decrease, lowering the rate of forward reaction.

Increasing Concentration of products

Increased concentration of the products accelerates the reverse reaction

The rate of a chemical reaction depends on the number of effective collisions between reacting molecules.

At equilibrium number of effective collisions for the forward and the reverse reactions are equal.

An increase in concentration of reactant increase the effective collision for the forward direction thus equilibrium shifted toward right by forming more products.

Formation of more product increase the effective collision for the reverse direction thus equilibrium shifted toward left by forming reactant and maintain the equilibrium.

2. Effect of change in pressure:

Equilibria that contain gases are influenced by changes in pressure.

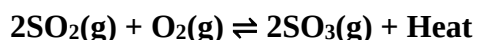
When pressure on a gaseous system at equilibrium increases, the system tends to reduce its volume to undo or minimize the effect of increased pressure.

This is done by decreasing the total number of gaseous molecules in the system. This is because, *at constant temperature and pressure, the volume of a gas is directly proportional to the total number of molecules of the gas present.*

Pressure $\propto$ 1/Volume

Volume $\propto$ total number of molecules

Consider the following equilibrium system. Which side contains smaller numbers of molecules?

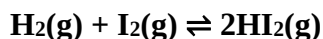


If we suddenly increase pressure on the system. The reaction will shift to the right, because in this direction three molecules (two of SO₂, and one of O₂.) react to produce two molecules (of SO₂), thus reducing the total number of gaseous molecules present.

Molecules with no effect of pressure:

There are certain equilibrium reactions in which the total number of molecules are same on either side.

For example



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When the pressure is changed in such a system, neither the forward nor the reverse reaction is favored because the number of molecules is the same on both sides.

3. Effect of change in temperature:

Chemical reactions that liberate heat are called exothermic.

The chemical reactions that absorb heat are called endothermic.

We can use Le-Chatelier's Principle to predict the direction of change. An increase in temperature adding heat favours the endothermic reaction and a decrease in temperature (removing heat) favours the exothermic reaction. Consider the following reaction.



According to the Le Chatelier's Principle an increase in temperature will shift the reaction from right to left in order to absorb the added heat and counteract the temperature increase.

As a result of this Change concentration of SO_3 , will decrease and concentrations of SO_2 , and O_2 , will increase.

4. Effect of catalyst:

A catalyst speeds up both, forward and reverse reaction to the same degree. So, catalyst has no effect on the equilibrium concentration of reaction mixture.

INDUSTRIAL APPLICATION OF LECHATELIER'S PRINCIPLE
Synthesis of Ammonia by Haber's process

The manufacture of ammonia by Haber's process is represented by the following equation.



This equation provides the following information.

- i) The reaction is exothermic.
- ii) The reaction proceeds with a decrease in number of molecules or moles.

Le Chatelier's principle suggests three ways to get maximum yield of ammonia.

i. Low Temperature

The forward reaction is exothermic therefore, low temperature will favour the formation of ammonia. The suitable temperature is 400°C .

**ii. High Pressure**

Since four molecules (one of N_2 and three of H_2) react to produce two molecules of NH_3 .

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Thus, high pressure will shift the equilibrium to the right side i.e. formation of NH_3 . The most suitable pressure is 200-300 atm.

Thus, the optimum condition for the equilibrium production of ammonia is low temperature and high pressure

Using Catalyst

A catalyst is used to increase the rate of reaction. Usually, a piece of iron with other metal oxides is used as the catalyst.

iii. Continual removal of ammonia

A final factor that greatly increases the production of ammonia is the continuous removal of ammonia as it is formed.

This is done by liquefying ammonia. The equilibrium mixture is cooled by refrigeration coils until ammonia condenses at -33.4°C and is removed.

N_2 and H_2 which do not liquefy at this temperature are recycled into the reaction chamber.

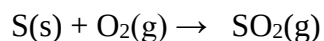
The stress caused by the continual removal of ammonia shifts the equilibrium toward the production of more ammonia.

Synthesis of H_2SO_4 by contact process

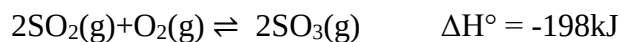
H_2SO_4 is produced by the contact process in the industry.

1. Formation of SO_2 :

SO_2 is produced by burning of S

**2. Oxidation of sulphur dioxide**

SO_2 is oxidized into SO_3 , using vanadium pentoxide (V_2O_5)



This reaction provides the following information.

- a) The reaction is exothermic.
- b) The reaction occurs with a decrease in number of moles.

Ways to maximize the yield of sulphuric acid using Le-Chatelier principle**a) Low temperature**

Since the forward reaction is exothermic, the low temperature favours the formation of sulphur trioxide. However, too low temperature slows down the reaction speed.

So, a compromised temperature of $450 - 500^\circ\text{C}$ is used.

b) High pressure

Since forward reaction produces fewer moles, high pressure shifts the equilibrium to the right. But maintaining high pressure is very expensive. So, the compromised pressure of 2 atm. is used.

c) Adding excess oxygen

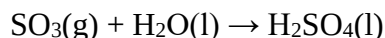
According to Le-Chatelier's principle, an increase in O₂ concentration at equilibrium shifts the equilibrium to the right.

However, adding too much O₂ decreases the concentration of SO₃, which slows down the reaction rate.

Thus, a degraded 1:1 ratio of SO₂ and O₂ is used to obtain high yields.

3. Conversion of SO₃ into H₂SO₄.

When SO₃ is formed, now its become easy to convert it into H₂SO₄ acid by reacting with H₂O.

**For More Information**

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